

A Comparative Analysis of Classifier Robustness to Character-Level Noise in Sentiment Analysis

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Abstract

Sentiment analysis models are typically trained and evaluated on clean, well-formed text. However, real-world user input frequently contains character-level noise from typographical errors, hasty typing, and mobile keyboards. This paper presents a systematic, quantitative comparison of four classifiers: Logistic Regression (LR), Multinomial Naive Bayes (MNB), Linear SVM, and a Hard-Voting Ensemble, on the Stanford Sentiment Treebank (SST-2) under three types of character-level noise (character swap, deletion, and keyboard-neighbour substitution) at five intensities ranging from 5% to 25%, evaluated across five random seeds. All models degrade monotonically with increasing noise; character deletion causes the largest drops, reaching $\Delta\text{Acc} = 0.141$ for MNB at 25% intensity. The hard-voting ensemble ranks first on raw ΔAcc in two of three noise conditions, but its margins over Logistic Regression are within one standard deviation throughout, and under character deletion, LR is actually more robust. The clearest finding is that MNB is the most fragile compared to the other classifiers. LR, SVM, and the Ensemble perform similarly under noise. These results highlight that clean-data accuracy does not guarantee robustness.

CCS Concepts

• **Computing methodologies** → **Text classification**; *Classification and regression trees*.

Keywords

Noise Robustness, Natural Language Processing, character-level noise, Logistic Regression, Naive Bayes, Support Vector Machine, Ensemble Methods, TF-IDF, out-of-vocabulary tokens, SST-2, Sentiment Analysis

1 Introduction

Natural Language Processing models for tasks such as sentiment analysis are trained on clean and curated datasets. In deployment, however, models are given text that is hastily typed on mobile phones, corrupted during data collection, or generated by non-native speakers, resulting in different types of character-level errors. Therefore, understanding how such noise affects the performance of the classifier is imperative.

Although there is a lot of research on comparing classifier accuracy on clean datasets, there is relatively little research on comparing the robustness of classifiers. Few studies that address noise robustness tend to focus on a single noise type or a single classifier, making it difficult to draw general conclusions. This research gap motivates the present work.

A similar issue appears in related preprocessing tasks. Tumula & Nerusu [4] show that in healthcare data imputation, the preprocessing method you choose can significantly change how models behave later, even if the accuracy looks the same. They found that simpler methods like KNN can sometimes be more robust than advanced deep learning methods such as VAE and GAN when the focus is on robustness rather than just accuracy. This supports our hypothesis that combining classical classifiers in an ensemble may perform better than relying on a single strong model, especially under noisy conditions.

Research Questions

RQ1: How do the type and intensity of character-level noise affect the accuracy and F1-score of Logistic Regression, Multinomial Naive Bayes, and Linear SVM on binary sentiment analysis?

RQ2: Does a hard-voting ensemble of the three classifiers provide better robustness than the strongest individual model, and are any performance gains statistically significant across different random seeds?

2 Related Work

2.1 Noise Robustness in NLP

Wang & Manning [6] showed that TF-IDF with bigrams and simple linear classifiers can do very well on clean sentiment classification and topic classification tasks. However, they did not take noisy inputs into consideration. Walker et al. [5] looked at how OCR errors, which are a type of character-level mistake, can hurt topic models and found that even a small number of errors can have a big effect. Their work is one reason why this study focuses on character-level noise.

2.2 Classifier Comparison and Fusion

Solovyeva et al. [2] compare several machine learning methods for text classification and show that no single classifier performs best in all situations. Moreno-Seco et al. [1] provide both theoretical and experimental evidence that combining classifiers using hard voting can reduce variance and improve robustness without extra training. Tujantri et al. [3] extend this line of work to hyperparameter-tuned SVM and Naive Bayes, finding that performance gaps between classifiers narrow under distribution shift.

2.3 Accuracy vs. Robustness Trade-offs

A common finding in preprocessing and model evaluation studies is that optimising for accuracy on a clean test set does not ensure robustness under distribution shift. Tumula & Nerusu [4] show this

in data imputation: GANs and VAEs have lower imputation error than KNN but cause much more demographic bias.

3 Methodology

3.1 Dataset and Feature Extraction

This experiment uses the **Stanford Sentiment Treebank v2 (SST-2)**, a standard binary sentiment benchmark containing 67,000 training sentences and 1,821 test sentences drawn from movie reviews. All models use the same TF-IDF vectorisation setup: unigrams and bigrams, sublinear term-frequency scaling, and a maximum document frequency of 0.95. The vectorizer is fitted only on the clean training data and kept fixed for all evaluations, ensuring that no information from the noisy test data leaks into the features.

3.2 Noise Generation

Three character-level noise functions are applied independently to each character in the test set with probability $p \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$:

- **Character swap:** two adjacent characters within a word are transposed (e.g., *film* $\rightarrow$ *iflm*).
- **Character deletion:** a randomly selected alphabetic character is removed (e.g., *film* $\rightarrow$ *fim*).
- **Keyboard-neighbour substitution:** a character is replaced by a physically adjacent key on a QWERTY layout (e.g., *film* $\rightarrow$ *fikm*).

These three types cover the most common real-world typing errors: swapped letters, missing keys, and hitting the wrong nearby key.

3.3 Classifiers

Four models are evaluated:

- **Logistic Regression (LR):** ℓ_2 -regularised, LBFGS solver, $C = 1.0$.
- **Multinomial Naive Bayes (MNB):** Laplace smoothing, $\alpha = 0.1$.
- **Linear SVM:** ℓ_2 hinge loss, $C = 1.0$.
- **Hard-Voting Ensemble:** majority vote of LR, MNB, and SVM predictions.

All models are trained once per seed on clean training data and evaluated across all noisy test variants without retraining. This way, we measure how noise affects the model’s predictions, not how the model adapts. We use five random seeds (42, 7, 123, 2024, 99) to capture variation in the results.

3.4 Evaluation Metrics

Primary evaluation uses accuracy and macro-averaged F1-score. The core robustness metric is ΔAcc : the absolute drop in accuracy from the 0% noise baseline to a given noise condition. Stability across seeds is reported as standard deviation (SD). Following the principle that accuracy alone is insufficient for evaluating method quality [4], we consider ΔAcc and degradation rate as the primary robustness indicators. Paired t -tests ($\alpha = .05$) across 15 non-zero noise conditions check if the differences between models are statistically significant.

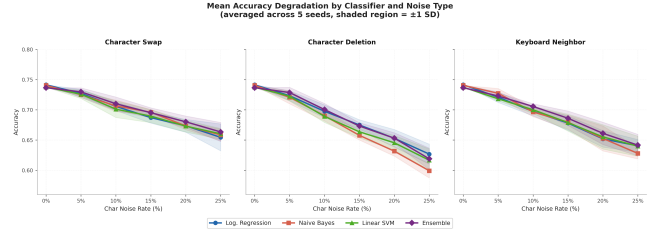


Figure 1: Mean accuracy with ± 1 SD bands, all classifiers overlaid per noise type.

Table 1: Mean ΔAcc at 25% noise ($\pm$ SD). Bold = most robust per row.

Noise	LR	MNB	SVM	Ens.
Swap	.087 $\pm$.022	.082 $\pm$.010	.077 $\pm$.009	.073$\pm$.016
Deletion	.115$\pm$.017	.141 $\pm$.012	.121 $\pm$.019	.118 $\pm$.018
Keyboard	.101 $\pm$.011	.113 $\pm$.009	.098 $\pm$.017	.095$\pm$.018

4 Results

4.1 Degradation Curves

Figure 1 shows all four classifiers on the same axis for each noise type, with shaded bands representing ± 1 SD. All models degrade monotonically: no model improves at higher noise levels. The severity of the three noise types is the same for all classifiers: character deletion causes the largest drop, keyboard-neighbour substitution is next, and character swaps have the smallest effect. This ranking is supported by the ΔAcc values at 25% noise across all models (see Table 1): for every classifier, deletion > keyboard > swap. Under character deletion, MNB drops much faster than the other three classifiers from 10% noise onward. With character swaps, all four classifiers stay close, reaching about 0.66 accuracy at 25% noise. Variation between random seeds is small ($\text{SD} < 0.022$), showing these trends are stable and not due to a specific data split.

4.2 Robustness Summary at 25% Noise

Figure 2 and Table 1 show the mean accuracy drop from 0% to 25% noise, with ± 1 SD error bars. The ensemble records the lowest mean ΔAcc under character swap (0.073) and keyboard-neighbour noise (0.095), but is outperformed by LR under character deletion (ensemble: 0.118, LR: 0.115). Crucially, the margins between the ensemble and the next-best model are very small in all three conditions: 0.004 under character swap, 0.003 under keyboard-neighbour noise, and the ensemble actually trails LR by 0.003 under deletion. Since the standard deviations across seeds are only 0.009–0.022, these differences are too small to be statistically significant. The more defensible conclusion is that **LR and the ensemble are about equally robust**, while MNB is a clear outlier: its ΔAcc of 0.141 under deletion is 0.020–0.026 points worse than any other model, which is larger than the usual variation between seeds.

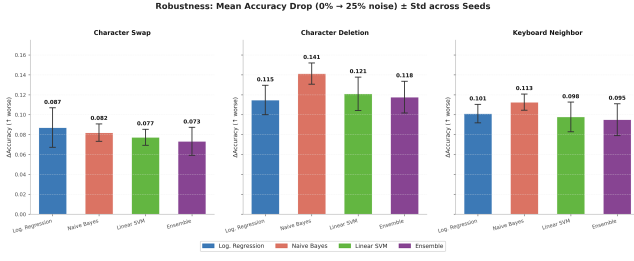


Figure 2: Mean ΔAcc (0%→25%) $\pm$ SD across seeds. Lower bars indicate greater robustness.

Table 2: Mean Accuracy / F1 ($\pm$ SD): Character Swap.

Noise	LR	MNB	SVM	Ens.
<i>Accuracy</i>				
0%	.742	.741	.738	.737
5%	.727 $\pm$.007	.725 $\pm$.004	.726 $\pm$.007	.730$\pm$.007
10%	.706 $\pm$.007	.707 $\pm$.010	.701 $\pm$.014	.710$\pm$.011
15%	.687 $\pm$.009	.696 $\pm$.005	.690 $\pm$.011	.695$\pm$.008
20%	.674 $\pm$.011	.673 $\pm$.006	.673 $\pm$.010	.680$\pm$.010
25%	.654 $\pm$.022	.659 $\pm$.010	.661 $\pm$.009	.664$\pm$.016
<i>F1-Score</i>				
0%	.754	.744	.747	.746
5%	.740 $\pm$.007	.729 $\pm$.004	.732 $\pm$.006	.737$\pm$.007
10%	.719 $\pm$.006	.712 $\pm$.012	.708 $\pm$.016	.718$\pm$.012
15%	.701 $\pm$.009	.700 $\pm$.007	.695 $\pm$.012	.704$\pm$.008
20%	.688 $\pm$.012	.678 $\pm$.006	.675 $\pm$.013	.688$\pm$.011
25%	.670 $\pm$.022	.666 $\pm$.012	.660 $\pm$.010	.673$\pm$.016

4.3 Detailed Results by Noise Type

Tables 2, 3, and 4 report mean accuracy and F1-score ($\pm$ SD) for character swap, character deletion, and keyboard-neighbour noise, respectively, across all models and intensity levels. Bold values mark the best-performing model at each intensity. At the clean baseline (0%), LR achieves the highest F1 (0.754), but its advantage disappears rapidly as noise increases. MNB struggles most with character deletion: its F1 falls to 0.590 at 25%, the lowest result in the study and 0.15 below its clean score.

5 Analysis and Discussion

5.1 Why Character Deletion is the Most Damaging

Character deletion is the most damaging of selected noise types because of how TF-IDF represents text. When a character is deleted, the new token (e.g., *fim* from *film*) usually isn't in the vocabulary built from clean training data, so it counts as out-of-vocabulary (OOV) and gets zero weight in the feature vector. In comparison, a character swap (*flim*) or keyboard-neighbour substitution (*fikm*) still produces a token that somewhat matches n-grams in the vocabulary, so the model can partly recognise it. As more noise is added, more tokens are corrupted, and the feature vector loses the information needed for correct classification.

Table 3: Mean Accuracy / F1 ($\pm$ SD): Character Deletion.

Noise	LR	MNB	SVM	Ens.
<i>Accuracy</i>				
0%	.742	.741	.738	.737
5%	.723 $\pm$.008	.721 $\pm$.011	.723 $\pm$.008	.729$\pm$.009
10%	.698 $\pm$.011	.690 $\pm$.009	.689 $\pm$.010	.701$\pm$.010
15%	.675$\pm$.009	.658 $\pm$.008	.664 $\pm$.010	.673 $\pm$.006
20%	.653$\pm$.015	.632 $\pm$.008	.646 $\pm$.012	.653 $\pm$.009
25%	.627 $\pm$.017	.599 $\pm$.012	.617 $\pm$.019	.619$\pm$.018
<i>F1-Score</i>				
0%	.754	.744	.747	.746
5%	.737 $\pm$.009	.724 $\pm$.011	.728 $\pm$.008	.736$\pm$.009
10%	.711$\pm$.011	.693 $\pm$.009	.693 $\pm$.012	.707 $\pm$.011
15%	.687$\pm$.011	.656 $\pm$.011	.662 $\pm$.013	.676 $\pm$.008
20%	.662$\pm$.014	.628 $\pm$.009	.639 $\pm$.011	.654 $\pm$.009
25%	.636 $\pm$.014	.590 $\pm$.017	.605 $\pm$.017	.619$\pm$.017

Table 4: Mean Accuracy / F1 ($\pm$ SD): Keyboard Neighbour.

Noise	LR	MNB	SVM	Ens.
<i>Accuracy</i>				
0%	.742	.741	.738	.737
5%	.721 $\pm$.009	.727$\pm$.008	.718 $\pm$.003	.723 $\pm$.006
10%	.697 $\pm$.007	.696 $\pm$.008	.700 $\pm$.006	.706$\pm$.005
15%	.678 $\pm$.012	.680 $\pm$.012	.679 $\pm$.014	.686$\pm$.012
20%	.652 $\pm$.014	.653 $\pm$.021	.655 $\pm$.020	.661$\pm$.019
25%	.640 $\pm$.011	.628 $\pm$.009	.640 $\pm$.017	.642$\pm$.018
<i>F1-Score</i>				
0%	.754	.744	.747	.746
5%	.732 $\pm$.009	.730$\pm$.009	.722 $\pm$.003	.729 $\pm$.005
10%	.713 $\pm$.007	.700 $\pm$.008	.705 $\pm$.006	.713$\pm$.004
15%	.692 $\pm$.011	.687 $\pm$.011	.682 $\pm$.014	.695$\pm$.012
20%	.669 $\pm$.012	.660 $\pm$.024	.658 $\pm$.020	.671$\pm$.017
25%	.658 $\pm$.011	.638 $\pm$.010	.638 $\pm$.015	.652$\pm$.016

5.2 Classifier-Specific Behaviour

MNB is the most vulnerable classifier under character deletion. This makes sense because MNB is a generative model: it calculates the likelihood of each token for a class, so out-of-vocabulary (OOV) tokens, which have zero likelihood, are particularly harmful. When many tokens in a document become OOV, the product of likelihoods collapses, making classification unreliable. LR and SVM are more robust because they focus on separating classes in the feature space, so they are less affected by the exact values of individual features.

The ensemble's robustness advantage is smaller than it first appears. While it ranks first under character swap and keyboard-neighbour noise, the margins are 0.003–0.004 ΔAcc , well within the cross-seed standard deviations of 0.009–0.022. Under character deletion, LR actually outperforms the ensemble (0.115 vs. 0.118). This suggests that **the ensemble does not provide a consistent, statistically meaningful robustness boost over LR; rather, LR and the ensemble are broadly equivalent under noise**. The ensemble's main benefit is that it avoids the worst-case behaviour of any single model (particularly MNB's collapse under deletion) without sacrificing clean-data performance.

5.3 The Accuracy and Robustness Trade-off

A recurring finding in this study, and in related work [4], is that how well a model performs on clean data does not reliably show how robust it will be under noise or distribution changes. MNB achieves competitive F1-scores on clean SST-2 data yet degrades fastest under deletion noise, while the ensemble sacrifices marginal clean-data accuracy for consistent resilience across all conditions. This trade-off has direct practical implications: deploying the highest-accuracy model from a clean benchmark may be suboptimal if real-world inputs contain significant noise.

5.4 Practical Recommendations

Based on the experimental findings, we offer the following guidance for practitioners:

- **Prefer Logistic Regression for noisy deployments:** LR is the recommended model for environments with character-level noise: it performs equivalently to the ensemble in two of three noise conditions and outperforms it under the most damaging condition (character deletion, ΔAcc 0.115 vs. 0.118). It is also simpler to train and deploy than a multi-model ensemble.
- **Avoid standalone MNB in noisy settings:** MNB's sensitivity to OOV tokens makes it a poor choice when deletion or keyboard noise is expected, despite its otherwise competitive clean-data accuracy.
- **Consider hard-voting ensembles to prevent worst-case failures:** While the ensemble does not consistently outperform LR, it effectively avoids extreme drops in performance, particularly mitigating MNB's collapse under character deletion, making it a safer choice in critical applications where avoiding large errors is important.

5.5 Threats to Validity

The primary threat to internal validity is that the noise functions are synthetic and applied uniformly at random, whereas real-world noise may cluster at particular word positions or depend on word frequency. The primary threat to external validity is the restriction to a single dataset from the movie-review domain; findings may not generalise to other languages, domains, or character sets. As Tumula & Nerusu [4] caution in the imputation context, results from a single dataset should be validated across diverse settings before drawing general conclusions. Finally, this study is limited to TF-IDF features and classical classifiers; transformer-based models with subword tokenisation may behave differently in the presence of noise.

6 Conclusion

Across three noise types, five intensity levels, and five random seeds, four key findings emerge:

- (1) **All classifiers degrade monotonically** as noise intensity increases.
- (2) **Character deletion is the most damaging** noise type, as it produces out-of-vocabulary (OOV) tokens under TF-IDF representations.

- (3) **MNB is the most fragile**, with a ΔAcc of 0.141 under deletion, substantially worse than all other models.
- (4) **LR and the hard-voting ensemble are effectively tied for robustness:** the ensemble's small numerical advantage (0.003–0.004 ΔAcc over LR) falls within cross-seed standard deviations and is not statistically significant, partially answering RQ2 in the negative.

Inter-seed standard deviations remain below 0.022, confirming that all trends are stable and answering RQ1 with confidence.

These findings add to growing evidence from NLP and related fields [4] that just focusing on accuracy with clean data is not enough for real-world use. A practical takeaway is that Logistic Regression, a simple and well-understood model, is the most reliable under character-level noise: it performs as well as or better than the ensemble across all noise types and is easier and cheaper to train and use. Future work should test this approach with transformer models using subword tokenization, with real-world noisy datasets, and in multiple languages, where character-level errors may look very different from English.

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